

Name: _____

Period: _____

Mixed Solving Practice

Solve the following questions using algebraic methods

1.

$$\begin{cases} a+b+c=10 \\ a-b-c=4 \\ a+b+2c=15 \end{cases}$$

$$7+b+5=10$$

$$b+12=10$$

$$\begin{array}{r} -12 \\ -12 \end{array}$$

$$b=-2$$

$$\begin{array}{r} a+b+c=10 \\ a-b-c=4 \\ \hline 2a=14 \\ \frac{2a}{2} \quad \frac{14}{2} \\ a=7 \end{array}$$

$$a=7$$

$$(7, -2, 5)$$

$$\begin{array}{r} a-b-c=4 \\ a+b+2c=15 \\ \hline 2a+c=19 \end{array}$$

$$2a+c=19$$

$$2(7)+c=19$$

$$\begin{array}{r} 14+c=19 \\ -14 \quad -14 \\ \hline c=5 \end{array}$$

$$c=5$$

2.

$$\begin{cases} x+y+z=18 \\ x-y+2z=25 \\ 5x-20y-2z=2 \end{cases}$$

$$8+1+z=18$$

$$9+z=18$$

$$z=9$$

$$(8, 1, 9)$$

$$-2(x+y+z=18)$$

$$x-y+2z=25$$

$$-2x-2y-2z=-36$$

$$6(-x-3y=-11)$$

$$6x-21y=27$$

$$-6x-18y=-66$$

$$\begin{array}{r} -39y=-39 \\ \frac{-39y}{-39} \quad \frac{-39}{-39} \\ y=1 \end{array}$$

$$y=1$$

$$x-y+2z=25$$

$$5x-20y-2z=2$$

$$6x-21y=27$$

$$-x-3(1)=-11$$

$$-x-3=-11$$

$$-x=-8$$

$$x=8$$

3.

$$\begin{cases} r-4s+5t=8 \\ r+2s-t=2 \\ r+s-2t=15 \end{cases}$$

$$r-7-2(-6)=15$$

$$r-7+12=15$$

$$r+5=15$$

$$r=10$$

$$(10, -7, -6)$$

$$-1(r-4s+5t=8)$$

$$r+2s-t=2$$

$$-r+4s-5t=-8$$

$$\begin{array}{r} 6s-6t=-6 \\ \frac{6s}{6} \quad \frac{-6t}{6} \quad \frac{-6}{6} \\ s-t=-1 \end{array}$$

$$s-t=-1$$

$$-s-t=13$$

$$-2t=12$$

$$\frac{-2t}{-2} \quad \frac{12}{-2}$$

$$t=-6$$

$$-1(r+2s-t=2)$$

$$r+2s-t=2$$

$$-r-2s+t=-2$$

$$-s-t=13$$

$$-s-(-6)=13$$

$$-s+6=13$$

$$\begin{array}{r} -s+6=13 \\ -6 \quad -6 \\ \hline -s=7 \end{array}$$

$$-s=7$$

$$s=-7$$

4.

$$\begin{cases} x + 2y - 3z = 9 \\ x - y - z = 4 \\ 3x + 6y - 9z = 27 \end{cases}$$

$$\begin{array}{r} (x + 2y - 3z = 9) \cdot 3 \\ \hline 3x + 6y - 9z = 27 \\ -3x - 6y + 9z = -27 \\ \hline 0 = 0 \end{array}$$

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5.

$$\begin{cases} x - y - z = 10 \\ 4x - 4y - 4z = -8 \\ x + y + 2z = 15 \end{cases}$$

$$\begin{array}{r} (x - y - z = 10) \cdot 4 \\ \hline 4x - 4y - 4z = 40 \\ -4x + 4y + 4z = -40 \\ \hline 0 = -48 \end{array}$$

No Solution

6.

$$\begin{cases} 2x - 7y + z = 11 \\ 6x - y - 5z = 93 \\ 3x - 14y - 10z = 6 \end{cases}$$

$$\begin{array}{r} -3(2x - 7y + z = 11) - 2(3x - 14y - 10z = 6) \\ \hline -6x + 21y - 3z = -33 \\ -6x + 28y + 20z = -12 \\ \hline \end{array}$$

$$\begin{array}{r} 20y - 8z = 60 \\ \hline 5y - 2z = 15 \quad \cdot 5 \\ 9y + 5z = 27 \quad \cdot 2 \\ \hline \end{array}$$

$$\begin{array}{r} 27y + 15z = 81 \\ \hline 9y + 5z = 27 \end{array}$$

$$25y - 10z = 75$$

$$18y + 10z = 54$$

$$\begin{array}{r} 43y = 129 \\ \hline 43 \end{array}$$

$$y = 3$$

$$2x - 7(3) + 0 = 11$$

$$2x - 21 = 11$$

$$2x = 32$$

$$x = 16$$

$$5(3) - 2z = 15$$

$$15 - 2z = 15$$

$$-2z = 0$$

$$z = 0$$

(16, 3, 0)